

Title: Second Largest Unique Element

Algorithm:

1. Initialize `largest` and `secondLargest` to minimum value.
2. Traverse the array.
3. If current element is greater than `largest`, update both variables.
4. Else if current element is less than `largest` and greater than `secondLargest`, update `secondLargest`.
5. If `secondLargest` remains unchanged, return `-1`.
6. Otherwise, return `secondLargest`.

Dry Run:

Use `[ 12, 35, 1, 10, 34, 1 ]` and show the updates of `largest` and `secondLargest`.

Output

34
-1

2.Title: Rotate Array K Steps to the Right

Algorithm:

1. Find the length of the array `n`.
2. Compute `k = k % n` to handle cases where `k > n`.
3. Reverse the entire array.
4. Reverse the first `k` elements.
5. Reverse the remaining `n-k` elements.
6. The array is now rotated to the right by `k` positions.

Dry Run:

Input: `[ 1, 2, 3, 4, 5, 6, 7 ], k = 3`

Original Array:

`[ 1, 2, 3, 4, 5, 6, 7 ]`

Reverse Entire Array:

[7, 6, 5, 4, 3, 2, 1]

Reverse First k=3 Elements:

[5, 6, 7, 4, 3, 2, 1]

Reverse Remaining Elements:

[5, 6, 7, 1, 2, 3, 4]

Output:

[5, 6, 7, 1, 2, 3, 4]

[3, 99, -1, -100]

Q3 · Strings

Title: Valid Anagram Check

Algorithm:

1. Convert both strings to lowercase.
2. Remove all spaces from both strings.
3. If lengths are different, return false.
4. Create a frequency map (array of size 256).
5. Traverse the first string and increment frequency count.
6. Traverse the second string and decrement frequency count.
7. Check the frequency map.
8. If any count is not zero, return false.
9. Otherwise, return true.

Dry Run:

Input:

"Listen", "Silent"

Convert to lowercase:

"listen", "silent"

Frequency Count:

listen → +1 for each character

silent → -1 for each character

All frequencies become zero.

Result:

true

Output:

true

true

false

Q4 · Strings

Title: Longest Substring Without Repeating Characters

Algorithm:

1. Initialize two pointers: `left = 0, right = 0`.
2. Use a hash map to store the last index of each character.
3. Traverse the string using `right`.
4. If the current character already exists in the current window:
 - Move `left` to `max(left, lastIndex + 1)`.
5. Update the character's latest index.
6. Calculate current window length.
7. Update maximum length if needed.
8. Continue until the end of the string.
9. Return the maximum length.

Dry Run:

Input:

"abcabcbb"

Right	Character	Window	Length
0	a	a	1
1	b	ab	2
2	c	abc	3
3	a	bca	3
4	b	cab	3
5	c	abc	3
6	b	cb	2
7	b	b	1

Maximum Length:

3

Output:

3

1

3

Q5 · SQL

Title: Top Earning Employees per Department

Algorithm:

1. Join Employees and Departments tables using `dept_id`.
2. Find the maximum salary in each department.
3. Match employees whose salary equals the department's maximum salary.
4. Return department name, employee name, and salary.
5. Employees with tied highest salaries are also included.

Query:

```
SELECT d.dept_name,
       e.name,
       e.salary
FROM Employees e
JOIN Departments d
ON e.dept_id = d.dept_id
JOIN (
    SELECT dept_id,
           MAX(salary) AS max_salary
    FROM Employees
    GROUP BY dept_id
) m
ON e.dept_id = m.dept_id
AND e.salary = m.max_salary;
```

Output:

dept_name	name	salary
HR	Alice	70000
IT	Bob	90000
IT	Charlie	90000

Q6 • SQL

Title: Monthly Order Revenue Report

Algorithm:

1. Select records from Orders where year is 2024.
2. Extract month from order_date.
3. Calculate:
 - Total revenue using SUM(amount)
 - Total orders using COUNT(order_id)
4. Group by month.

- Sort results by total revenue in descending order.

Query (MySQL):

```
SELECT DATE_FORMAT(order_date, '%b') AS month,
       SUM(amount) AS total_revenue,
       COUNT(order_id) AS total_orders
FROM Orders
WHERE YEAR(order_date) = 2024
GROUP BY MONTH(order_date),
         DATE_FORMAT(order_date, '%b')
ORDER BY total_revenue DESC;
```

Output:

month	total_revenue	total_orders
Mar	15000	120
Jan	12000	95
Feb	10000	80

Q8 · API Integration

Title: Weather Summary with Conditional Logic

Algorithm:

- Choose a city and note its latitude and longitude.
- Send a GET request to Open-Meteo API.
- Parse the JSON response.
- Extract:
 - temperature
 - weathercode
- Use a dictionary/map to convert weather code into a readable condition.
- Display city name, temperature, and weather condition.

7. Handle request errors gracefully.

Final Output:

City : London
Temperature: 18.2°C
Condition : Overcast

Output:

City : London
Temperature: 18.2°C
Condition : Overcast